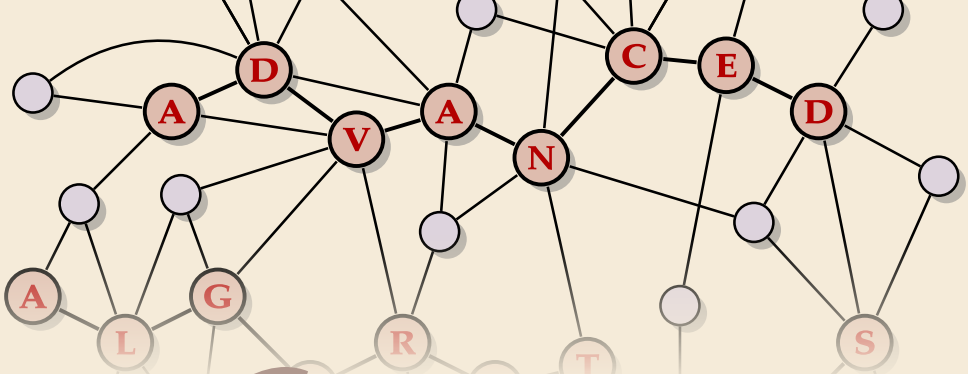


6 Advanced Parameterized Ideas

Advanced Algorithms · Winter 2026–27

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CS627 (Winter 2026–27)
Philipps-Universität Marburg
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Outline

6 Advanced Parameterized Ideas

6.1 Linear Programs – A Mighty Blackbox Tool
6.2 Linear Programs – Reformulation Tricks
6.3 Linear Programs – The Simplex Algorithm
6.4 Integer Linear Programs
6.5 LP-Based Kernelization
6.6 ETH-Based Lower Bounds

6.1 Linear Programs – A Mighty Blackbox Tool

Linear Programs

- ***Linear programs (LPs)*** are a class of optimization problems of **continuous** (numerical) variables
- can be exactly solved in worst case polytime (LinearProgramming $\in$ P)
  - interior-point methods, Ellipsoid method
- routinely solved in practice to optimality with millions of variables and constraints
  - Simplex algorithm, interior-point methods
  - many existing solvers, commercial and open source (e. g., HiGHS)

Hessy James's Apple Farm

- Hessy tries to maximize the profit of his apple farm
  - He is committed to promote regional Hessian heirloom varieties, so he only grows *"Sossenheimer Roter"* and *"Korbacher Edelrenette"*
  - each tree of *"Sossenheimer Roter"* yields apples worth € 195 per year
  - each tree of *"Korbacher Edelrenette"* yields applies worth € 255 per year
  - He has an orchard of 5 000 m^2
  - each tree needs 4 m^2 of orchard space
  - each tree of *"Sossenheimer Roter"* needs 6 kg of organic fertilizer and 1 h harvest effort per year
  - each tree of *"Korbacher Edelrenette"* needs 4.5 kg of organic fertilizer and 3 h harvest effort per year
  - Hessy can only afford 3000 kg of fertilizer and 1700 h of harvester time per year

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⇝ How many trees of each variety should Hessy plant?

- What will constrain us most? Space? Fertilizer? Harvest hours?
- What profit can Hessy expect?

Formal Linear Program for Hessy James's Apple Farm

- Classic application of linear programming in *operations research (OR)*
- We formally write LPs as follows:

optimization goal

objective function

constraint

$$
\begin{aligned}
&\textbf{Maximize:} && 195s + 255k && && \\
&\textbf{Subject to:} && 4s + 4k && \leq 5000 && \text{(Orchard constraint)} \\
& && 6s + 4.5k && \leq 3000 && \text{(Fertilizer constraint)} \\
& && 1s + 3k && \leq 1700 && \text{(Harvest constraint)} \\
& && s && \geq 0 && \text{(Non-negativity)} \\
& && k && \geq 0 && \text{(Non-negativity)}
\end{aligned}
$$

name of the LP

(P)

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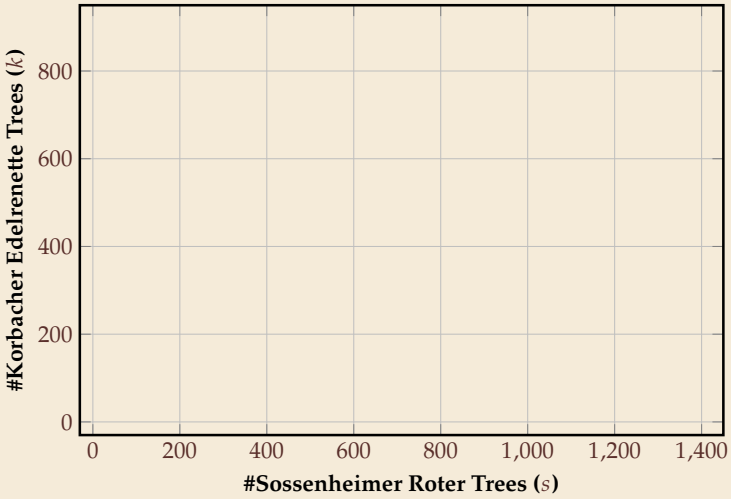
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\end{aligned}
\qquad (P)
$$

name of the LP

- **Terminology:**
  - s and k are the two *variables* of the problem; these are always real numbers.
  - A vector $(s, k) \in \mathbb{R}^2$ is a *feasible solution* for the LP if it satisfied all constraints.
  - The largest value of the objective function (over all feasible solutions) is the *(optimal) value* z^* of the LP
  - A feasible solution $(s^*, k^*) \in R^2$ with optimal objective value z^* is called an *optimal solution*

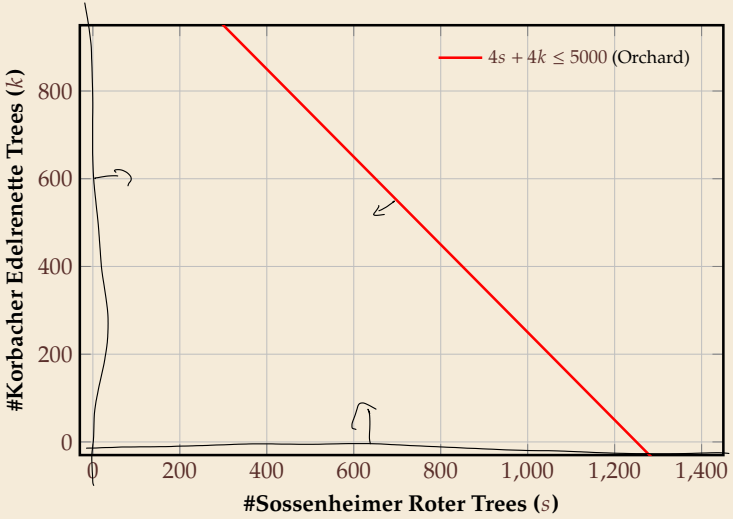
2D LPs – Graphical Solution

LPs with **two** variables can be solved graphically



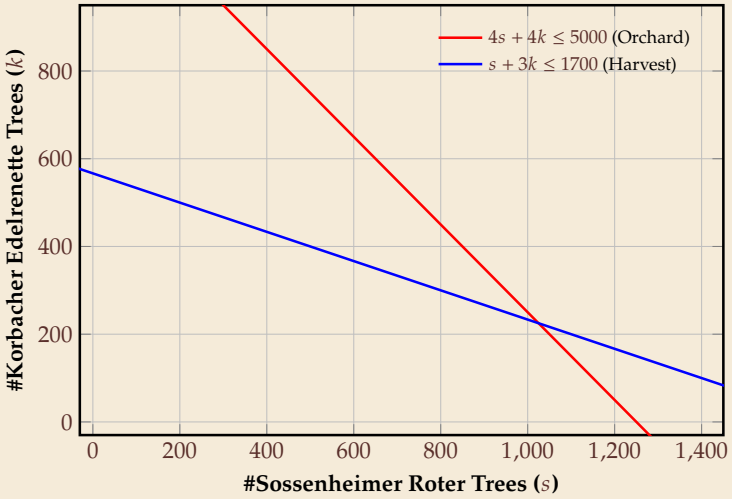
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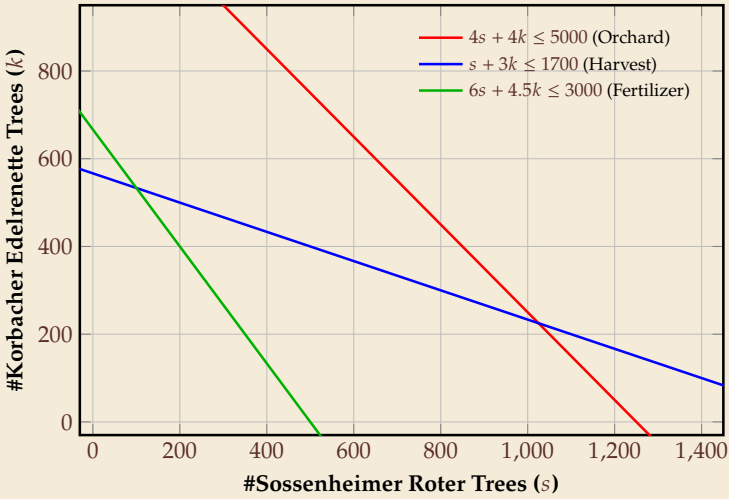
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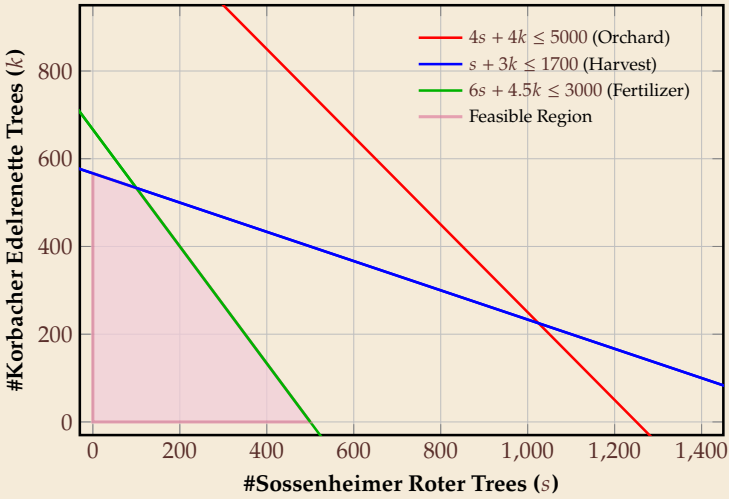
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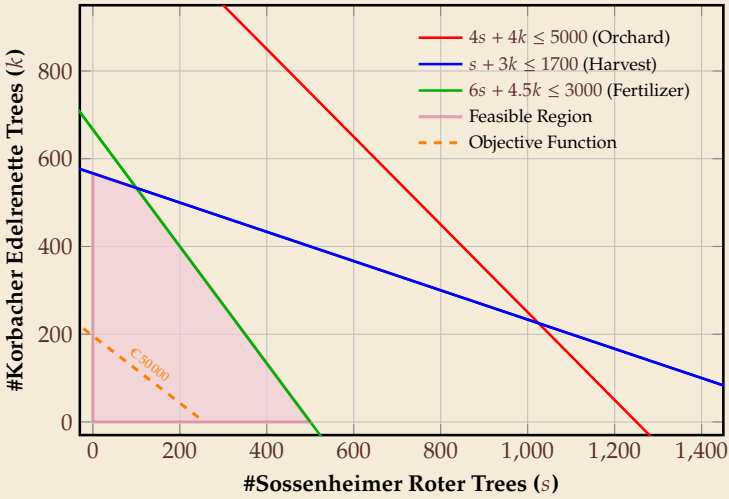
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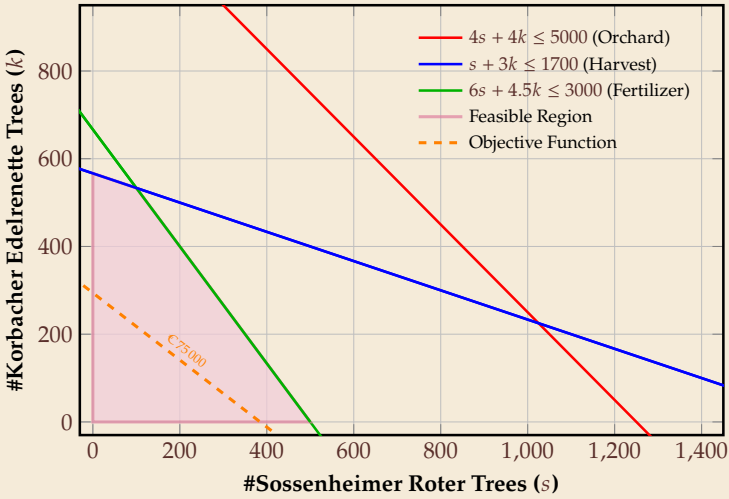
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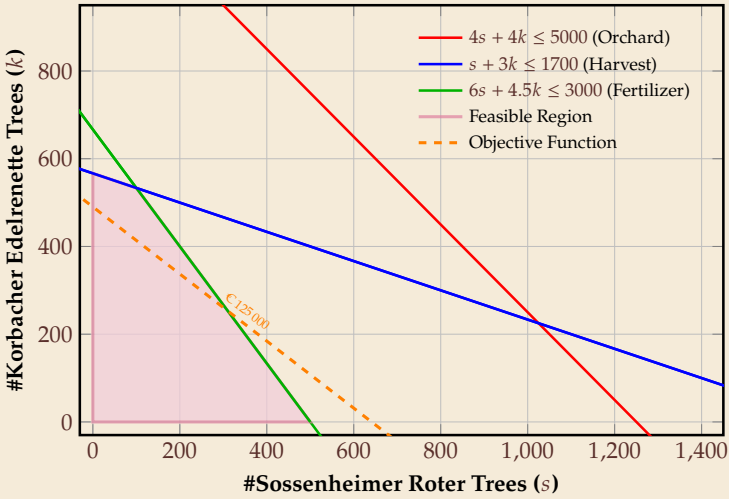
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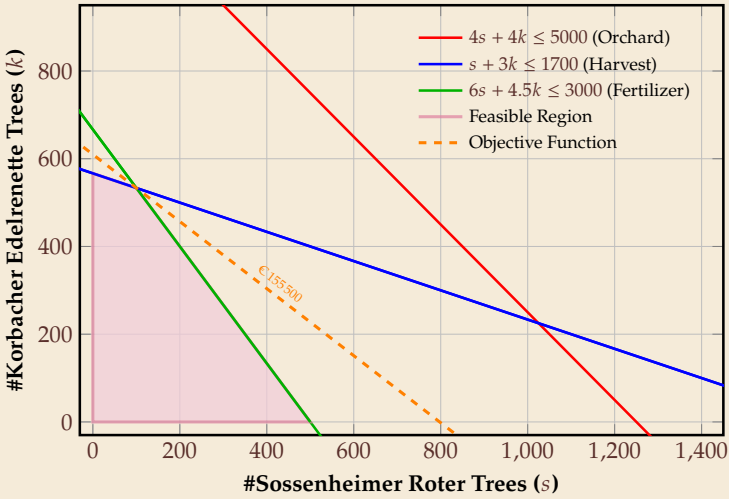
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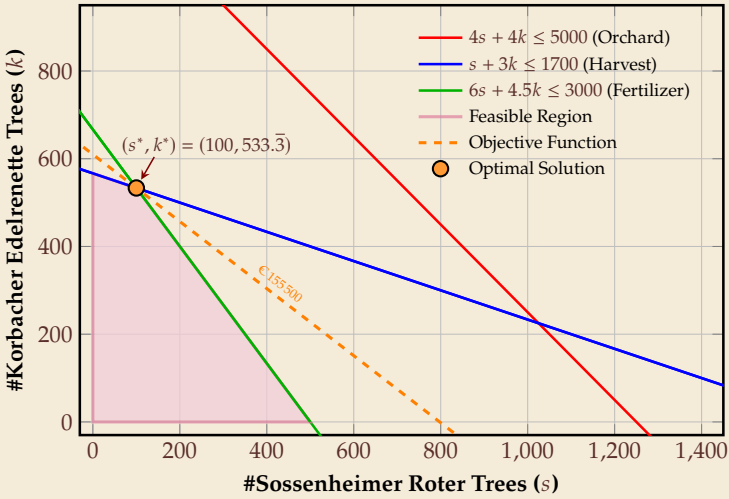
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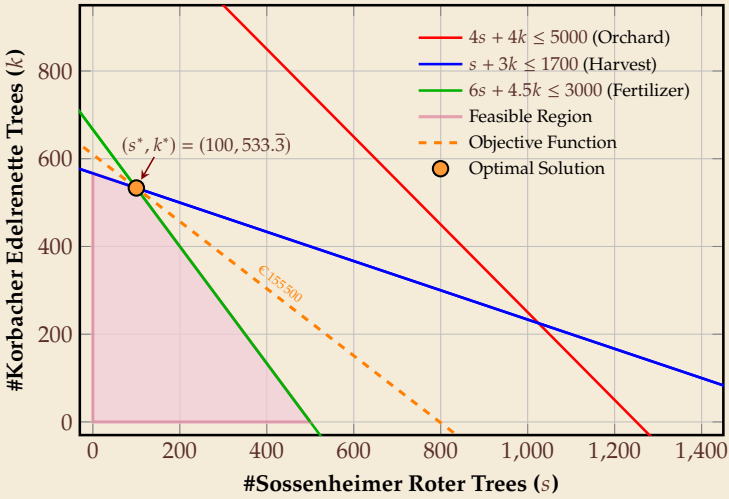
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2D LPs – Graphical Solution

LPs with **two** variables can be solved graphically



- ⇝ Hessy should plant
  - 100 *Sossenheimer Roter* trees and
  - $533+\frac{1}{3}$ *Korbacher Edelrenette* trees (hmm . . .)
  - Harvest **and** fertilizer *tight*
  - orchard space isn't
- ⇝ know what to change